Reviewer 2

This study demonstrates a land cover classification methodology focused on tropical peatland areas. The various experiments are implemented in Google Earth Engine and based on a random forest algorithm and using S1, S2 and DEM data. Six different feature combinations (consisting of bands, indices, temporal features and texture features) were applied and assessed in a small study area and the best performing combination was used to apply the methodology on a broader area.

The study is well written and structured. The topic, the methods and the results are clearly presented, in general.

General comments and suggestions for authors:

1. Since you are using DEM data in your study, it is important to provide information on the topography of the study area in section 2.1.

   RESPONSE: Thanks for the suggestion. Topographic detail has been added to section 2.1, as recommended.
2. Please provide the preprocessing level of the S2 data in the text as well (L1C) besides Table 1. Why weren't the L2A level S-2 products used instead? Why weren't any atmospheric corrections applied to the S2 data?

   RESPONSE: Thanks for the comments. The pre-processing level of the S2 data has been provided in the second paragraph of section 2.2. Unfortunately, insufficient cloud free L2A_SR data were available to create a composite image, so it was necessary to use L1C data. Also, using L1C data enabled us to apply a cloud masking function which relied on bands B1 and B10 (not available in L2A_SR) which led to a cloud free composite.

3. What is the red box in the figure 1? In combination with figure 4 it gives the impression of a test area, where the methodology was tested and a broader area, where the methodology was applied. If this is the case, name your two study areas (eg "test area" and "study area") so it can be clear from the beginning of the paper. Also, clearly name the areas in the legend of Figure 1. In regard to this, it must also be clearly stated in the text, from which study area the reference samples came from.

   RESPONSE: Thanks for the comments. For clarity, the study area map has been modified to eliminate the red box in figure 1 which suggested a test area separate within the study area. In figure 4 we merely seek to present a zoomed in view of a small part of the study area for ease of viewing by the reader.

4. I believe Table 7 illegible to the reader. Perhaps this table could be turned into a graph, similar to Figure 6, with a separate graph for each class. Or maybe colors could be added to highlight the large values in the table.

   RESPONSE: Thanks for the comment. Tables 7 has been altered as instructed; shading has been added to highlight the large values.

Comments by line:

141 What is the red box in the figure? Please delineate the exact study area boundaries or if the red box represents them, add it to the legend of the map accordingly.

RESPONSE: The study area map has been modified to eliminate the red box in figure 1.

168 Why is "undefined peatland boundaries" considered a threat? This statement is too general.

RESPONSE: This statement has been removed.

195 The creation of only one composite S2 image for the whole year would produce errors due to the changes in seasons. Why was there only one composite created instead of two (eg. one for the wet season and one for dry season)?

RESPONSE: Unfortunately, it was not possible to create a cloud free composite image during the rainy season; hence the approach used in our study. We therefore worked to minimize any seasonal influence by computing temporal features based on soil moisture conditions from the Sentinel-1 imagery, and also via our compositing approach.

196 What benefit did the mean composite of the 40-60 percentile range offer over the mean composite of the full range of values?

RESPONSE: We used a central percentile (40-60) to minimize the effects of extreme values on image compositing. Unlike the full range of values, the 40-60 percentile range is less variable, more temporally stable and captures a more representative spectral characterization of land cover by minimizing extreme atmospheric effects – given that we used a Level 1C product – and the challenge of cloud persistence. Further, this approach helps minimize errors due to seasonal and phenological variations.

215 Were any GIS techniques applied to achieve uniformity of reference data? Was any minimum distance used?

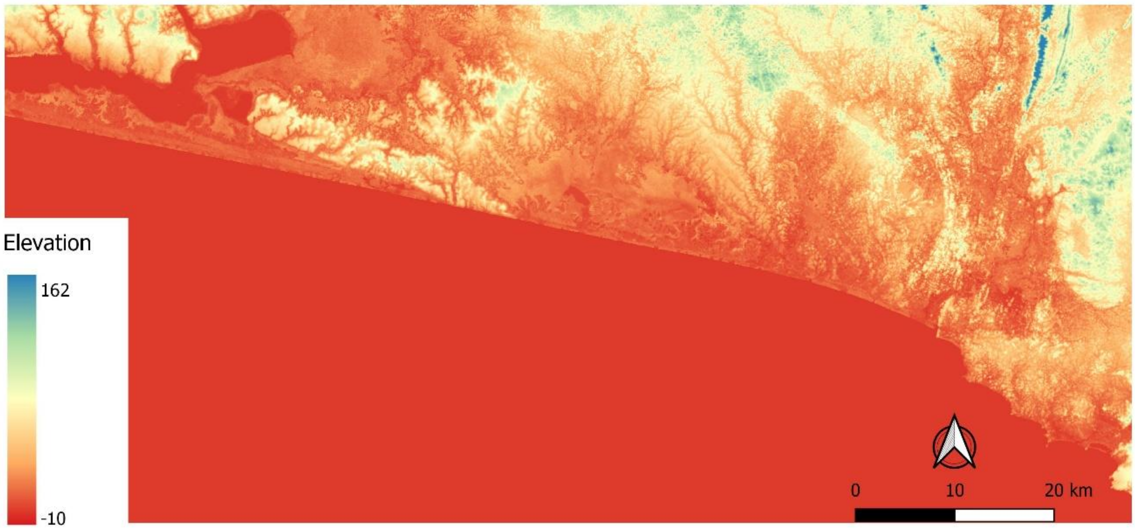
RESPONSE: We applied a stratified random sampling approach. Though no minimum distance rule was applied, stratification helped to capture spatial and spectral landscape variations while minimizing autocorrelation. This point has been clarified in the text.

227 Since training samples were collected as polygons and test samples in the form of points, in what unit are the reference samples presented in the table? Pixels, polygons?

RESPONSE: The polygon extents were used to extract image pixels to train the classifier. The table has been edited to make this clear.

430-432 It is evident from the accuracy metrics in table 6 that the inclusion of the DEM-derived information improved the discrimination of high-vegetation classes (forest, coconut, rubber, palm, sparse) as well as built-up areas. This indicates that the DEM information may work more as a DSM information rather than true topography information. This should be highlighted in the discussion section.

RESPONSE: This is a very interesting observation that is worthy of investigation, but a full study of how SRTM data represent the Earth’s surface is perhaps beyond the scope of this paper. We have rechecked this issue and we believe that the SRTM data are functioning effectively as a DEM as first thought. On inspecting our results, it is clear that the land cover classes are influenced strongly by topography. For instance, rubber plantations and natural forest are generally distributed on hilly land, while coconut plantations are found in low-lying coastal areas. Field survey corroborates these interpretations. Below we show the SRTM DEM which indicates topography throughout the study area.



SRTM data of the study area showing estimated elevation above sea level (m).

430 Important variables may occur in RF because a feature (eg elevation) might have considerable differences in its value range. Were all the features rescaled to the same range before performing variable importance?

RESPONSE: All input features were standardized by subtracting the mean and scaling to a unit variance. This detail has been added in the text to give more clarity.

435-437 It might be better for this sentence to be moved in the discussion section.

447-449 It might be better for this sentence to be moved in the discussion section.

451-454 It might be better for this sentence to be moved in the discussion section.

456-457 It might be better for this sentence to be moved in the discussion section.

468-474 It might be better for this sentence to be moved in the discussion section.

477-491 It might be better for this sentence to be moved in the discussion section.

RESPONSE: Thank you for the recommendations. Most of these sentences have been moved as recommended, though in one or two cases they did not fit well in the discussion session.